

Note that the economist Robert Lucas dealt a blow to econometrics by arguing that if people were rational then their rationality would cause them to figure out predictable patterns from the past and adapt, so that past information would be completely useless for predicting the future (the argument, phrased in a very mathematical form, earned him a Nobel Memorial Prize in Economics). We are human and act according to our knowledge, which integrates past data. I can translate his point with the following analogy. If rational traders detect a pattern of stocks rising on Mondays, then, immediately such a pattern becomes detectable, it would be ironed out by people buying on Friday in anticipation of such an effect. There is no point searching for patterns that are available to everyone with a brokerage account; once detected, they would be ironed out.

Somehow, what came to be known as the *Lucas critique* was not carried through by the "scientists". It was confidently believed that the scientific successes of the industrial revolution could be carried through into the social sciences, particularly with such movements as Marxism. Pseudoscience came with a collection of idealistic nerds who tried to create a tailor-made society, the epitome of which is the central planner. Economics was the most likely candidate for such use of science; you can disguise charlatanism under the weight of equations, and nobody can catch you since there is no such thing as a controlled experiment. Now the spirit of such methods, called *scientism* by its detractors (like myself), continued past Marxism, into the discipline of finance as a few technicians thought that their mathematical knowledge could lead them to understand markets. The practice of "financial engineering" came along with massive doses of pseudoscience. Practitioners of these methods measure risks, using the tool of past history as an indication of the future. We will just say at this point that the mere possibility of the distributions not being stationary makes the entire concept seem like a costly (perhaps *very costly*) mistake. This leads us to a more fundamental question: the problem of induction, to which we will turn in the next chapter.

SEVEN



THE PROBLEM OF INDUCTION

On the chromodynamics of swans. Taking Solon's warning into some philosophical territory. How Victor Niederhoffer taught me empiricism; I added deduction. Why it is not scientific to take science seriously. Soros promotes Popper. That bookstore on 21st and Fifth Avenue. Pascal's wager.

From Bacon to Hume

Now we discuss this problem viewed from the broader standpoint of the philosophy of scientific knowledge. There is a problem in inference well known as the problem of induction. It is a problem that has been haunting science for a long time, but science has not been as harmed by it as the financial markets. Why? Because the randomness content compounds its effects. Nowhere is the problem of induction more relevant than in my world of finance – and nowhere has it been as ignored!